

The potential for ground-mounted photovoltaics in Wallonia

Brownfields, activity zones, water bodies and agrivoltaics under the Walloon rules, and what it implies for the capacity limit used in PyPSA-Wal

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Abstract

PyPSA-Wal caps Walloon ground-mounted photovoltaics at 13 000 MW, the sum of published estimates for brownfields and agrivoltaics. PyPSA-Eur’s own land analysis, which knows no Walloon rule, gives 37 GW for the same node. This report computes the potential from the Walloon Region’s own instruments — the plan de secteur and the CoDT, the *circulaire relative aux permis d’urbanisme pour le photovoltaïque* of 14 March 2024, the Natura 2000 network and the nature reserves, the landscape and heritage perimeters, the flood-hazard and karst maps, the 2024 declared agricultural parcels (SIGEC), the WALOUS 2023 land-cover map and the 2021–22 LiDAR terrain model — on a 20 m grid covering the whole Region, and tests the constraints against the 88 ground-mounted parks mapped in Wallonia.

Unlike wind, Wallonia has *no* numeric framework for ground-mounted PV: the circular is “un outil d’aide à la décision” with lists of sites to refuse and to avoid, and no setback, size, slope or farm-share figure. The answer is therefore organised in two layers. The *technical* layer is the land that survives every exclusion and is large and wide enough for a park, classified by what it is. The *policy* layer is which of those categories a rule opens, and to what share.

The land is not the constraint. 6 275 km² survive the exclusions and the park rule (37.1 % of the Region), a technical potential of 356 GWc. Almost all of it is farmland: the declared agricultural parcels alone carry 261 GWc on 533 133 ha, 72 % of the utilised agricultural area.

The reference is 6.3 GWc, in three parts. What the circular opens without derogation: brownfields, landfills, spoil tips, the unexploited parts of extraction zones, verges wide enough for a park, industrial basins and quarry lakes, and a quarter of the free land of the activity and public-service zones. The derogation it admits on soils of demonstrated poor quality, taken at the share that could realistically be demonstrated: 5.1 % of the mapped-poor, unfarmed land (140 MWc). And agrivoltaics on grassland at the Livre blanc’s 0.6 % of the agricultural area (3.1 GWc) — a figure the Livre blanc derives for 2030, read here as a policy already on the table for a model whose horizon is 2050. The circular strictly as it stands, without derogation or agrivoltaics, gives 2.6 GWc.

The poor soils are mostly not poor. The only regional map ranks soils by percentiles of plant-available water and depth; its lowest classes are stony loams that farmers crop wherever they can (89 % of the land available to them, more than the average soil). Removing the land in the LPIS register and the gardens and farmyards, and weighting what is left by the revealed agricultural use of its soil unit, leaves 241 ha of 4 697.

The number is decided by the farmland rule. Raising the agrivoltaic quota to 1 % gives 8.3 GWc, to 2 % 13.5; every eligible grassland parcel 126.0; every eligible parcel 252.0. Every other choice moves the reference by a few GWc at most (§6).

For the model: the 13 GW cap is not a land limit but a policy assumption: it is what the reference gives with agrivoltaics on 2 % of the agricultural area instead of 0.6 % (§8).

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1 The question, and the scope of the answer

1.1 The number under review

PyPSA-Wal’s ground-mounted PV technology, `solar`, is capped in the Walloon node at 13 000 MW for every horizon (`data/walloon/custom_potentials.csv`), sourced as “Cluster TWEED, Livre blanc de l’agrivoltaïsme (2025), Energie Commune, Renouvelable (2025)” and described as the sum of the brownfield and the agrivoltaic potential. Rooftop PV and car-park canopies are a separate technology, `solar rooftop`, capped at 46 GW. This study is about the first number only: parks on the ground, including floating parks, and agrivoltaics.

1.2 What PyPSA-Eur would have said

PyPSA-Eur derives a land-based potential for every node from CORINE land cover and Natura 2000, at a fixed capacity per unit of available land. For the Walloon node it gives 37 GW of PV (`docs/renewable-potentials.md`, §6). That calculation admits all agricultural land at a small share and ignores the plan de secteur, the protected landscapes and the Walloon policy on farmland; it is a bound on land, not a potential.

1.3 Scope

The administrative Walloon Region (16 905 km²), as for the onshore-wind study. Out of scope: rooftops, façades and car-park canopies (the model’s `solar rooftop`), grid capacity, costs, and the question whether a potential will be built.

2 The Walloon rules for ground-mounted PV

2.1 No numeric framework

Wind has a *cadre de référence* adopted by the Government, with setbacks, park sizes and landscape rules. PV has a ministerial circular: the *circulaire relative aux permis d’urbanisme pour le photovoltaïque* of 14 March 2024 (Minister W. Borsus, Moniteur belge of 16 April 2024), which replaced that of 12 January 2022. It describes itself as “un outil d’aide à la décision à destination des différentes autorités compétentes” and states its line: “privilégier l’installation intégrée dans le paysage et qui n’a pas d’impact sur l’occupation du sol”. It contains no setback from dwellings, no minimum or maximum size, no height, slope or farm-share figure. Every number of this study that is not a legal perimeter is therefore labelled as what it is: a technical criterion, a modelling judgement or a policy choice, and each is priced in §6.

2.2 What the circular refuses and what it asks to avoid

- **Refuse** (p. 9): “les sites protégés (réserve naturelle, réserve forestière, zone humide d’intérêt biologique, cavité souterraine d’intérêt scientifique, site Natura 2000)”. Implemented as exclusions.
- **Avoid**: the SGIB, “les plans d’eau (hormis les bassins industriels)”, the *périmètres de liaison écologique*; “des paysages protégés (périmètres d’intérêt paysager, de point de vue remarquable, d’intérêt culturel, historique et esthétique, sites classés, etc.)”; zones of “risque naturel ou une contrainte géotechnique majeurs, comme le karst”. Implemented as exclusions (the reference reads “éviter” as a rule, as a permitting authority would), except the SGIB and the remarkable viewpoints, which have no usable public geometry (§9).
- **Farmland** (p. 5): “la terre agricole, entendue comme terrain utilisé pour l’agriculture quelle que soit son affectation au plan de secteur, doit remplir sa vocation première qui est d’être nourricière”. Research-linked pilot projects may be admitted where “la nouvelle fonction

énergétique s'exerce d'une façon qui soit compatible avec la fonction agricole". Otherwise: "ne pas utiliser de parcelles agricoles en cours d'exploitation".

- **Economic-activity zones:** "les installations photovoltaïques ont leur place pour autant qu'elles ne mettent pas en péril la destination première de la zone".
- **Brownfields:** to be brought back to economic activity first; "en attendant leur réhabilitation, les friches industrielles peuvent accueillir, à titre temporaire ... des champs photovoltaïques".
- **Preferred sites** (p. 5): "des sites dégradés et/ou non utilisés" — illustrated by an industrial basin, a former marshalling yard, a closed landfill and a road verge.

2.3 The plan de secteur and the CoDT

The CoDT (version applicable from 1 September 2026) names no zone in which a grid-connected PV park is an admitted use — wind turbines, by contrast, are named in three. Conformity follows from the general destination of each zone, and the circular reads it as follows: conform or tolerated in the zones d'activité économique mixte and industrielle, the zones de services publics et d'équipements communautaires (a PV park producing "énergie destinée exclusivement à la collectivité" is an activity of general interest, D.IV.22 al. 1 7° k), and for a limited time in the parts of extraction zones and landfill zones not yet exploited (D.II.33, D.II.41, D.II.26 §2, D.IV.80). Not admitted: the agricultural zone (except modules feeding a building on the same property, D.II.36 §2, R.II.36-11), the forest zone, the green zones, the zones naturelles and the parks. A derogation is possible (D.IV.11, with the three cumulative conditions of D.IV.13), which the circular asks to read "de manière restrictive", noting that in agricultural zone it can concern "des biens déjà artificialisés ou impropres à la fonction agricole, tels que des parkings, des décharges, ou encore des terres où il est démontré que la qualité agronomique du sol est médiocre".

2.4 Agrivoltaics

No decree or order on agrivoltaics had been adopted in September 2026. The Government's regional policy declaration (DPR 2024–2029) announces a revision of the renewable acceleration-zone decree to exclude "le photovoltaïque dans les parcelles agricoles" from it, and to "encadrer strictement" renewables in agricultural zone. The *Livre blanc de l'agrivoltaïsme en Wallonie* (Cluster TWEED, July 2024, completed January 2025), signed by the farmers' federation FWA, ULiège–Gembloux, the DSOs and the renewable sector, is the reference position of the sector. It recommends that an agrivoltaic project be "prioritairement un projet de production agricole", a yield maintained at 80 % at least, a ground coverage of at most 40 % for pilots, no destruction of permanent grassland, and it estimates that the PV still to be built to 2030 (2 916 MWc) would take about 4 483 ha, 0.6 % of the agricultural area, if all of it were agrivoltaic. It proposes a quota of agrivoltaic capacity at 20 % of rooftop capacity. Its GLAES-based mapping finds 56 787, 145 855 and 245 233 ha suitable in three scenarios of increasing breadth. The only agrivoltaic field built in Wallonia, Wierde, has 10 MWc on 14 ha with sheep.

3 Method

3.1 The workflow

A stand-alone Snakemake workflow, `workflow_pv_wal/`, retrieves every layer live from the Géoportail de la Wallonie (CC-BY 4.0) and OpenStreetMap, burns each one on its own onto a single 20 m grid in EPSG:3035 (a cell is inside a layer when its centre is), and evaluates every scenario from those burns (`scripts/pv_lib.py`). 20 m rather than the 100 m of the wind study: a PV park is a few hectares and the median declared parcel about one and a half. Every figure of this report, of its tables and of the dashboard comes from the same evaluation.

3.2 Exclusions

Seven families are applied in a fixed order, then water; a cell removed by several is counted in the first (Table 3):

1. **nature**: Natura 2000, nature reserves (domaniales and agréées), forest reserves, ZHIB, CSIS — the circular’s refusals — plus the zones naturelles and parks of the plan de secteur, the *périmètres de liaison écologique*, and the spoil tips still in place, which are among the richest dry grasslands of the Region;
2. **forest and green zones** of the plan de secteur;
3. **woodland** anywhere: a 20 m cell more than 50 % covered by trees over 3 m in WALOUS 2023 (a PV park there needs a clearing permit, for which the circular gives no ground);
4. **landscape and heritage**: *périmètres d’intérêt paysager* and *d’intérêt culturel*, historique ou esthétique of the plan de secteur, classified sites, ensembles and their protection zones, UNESCO buffer zones;
5. **hazards**: high flood hazard (overflow), strong karst constraint, landslide and rock-fall constraints, slopes over 30°, catchment protection zones IIa;
6. **terrain**: slope above 15 %, and north-facing slopes (aspect 315°–45°) above 7 %, from the LiDAR terrain model averaged to 20 m;
7. **built**: a cell 20 % or more covered by buildings, or 50 % or more sealed (roads, car parks, railways) in WALOUS 2023, and the footprint and verge of plan-de-secteur roads, railways and HV lines;

then water (a PICC water body of 1 ha or more, or a cell half covered by water), which is kept for floating PV. The “industrial basins” are the PICC artificial basins below 100 ha — settling ponds, the Coo pumped-storage basins — and every water body inside an extraction zone, an operating quarry or a brownfield (the quarry lakes). The one larger PICC basin, the Plate-Taille reservoir of the Eau d’Heure (348 ha, hydro, drinking water and leisure), is not an industrial basin and counts with the other water bodies.

3.3 Gisements: what the surviving land is

Each surviving cell is assigned to the first gisement whose test it passes. The order encodes the circular’s reading of farmland: a parcel declared to the SIGEC in 2024 is farmland whatever its zoning, so it comes before the extraction and activity zones — a field in an extraction zone is a field. The degraded categories come first only because a brownfield, a landfill or a spoil tip is not farmed. The gisements are listed in Table 4.

3.4 From land to parks, and to capacity

A **park rule** turns land into sites: a morphological opening removes every strip narrower than 60 m and every contiguous piece smaller than 1 ha is dropped. The circular asks to “éviter l’éparpillement de petites installations au sol” and prefers grid-injecting parks “d’une superficie importante”, without a figure; both thresholds are judgements and are priced. Capacity is eligible area times a density per system, set against the Walloon parks (§4): 1.00 MWc/ha on the ground (bracket 0.70–1.40), 0.70 for agrivoltaics on grassland (Wierde: 0.71), 0.35 on arable crops (vertical bifacial rows or wide-row canopies), 0.60 for canopies over fruit, and 1.50 MWc per hectare of floats on 25 % of a water body.

3.5 The policy layer

Which gisements are open, and to what share, is the policy. The **reference** opens brownfields (temporarily, as the circular admits), landfills, levelled spoil tips, the non-exploited extraction zones, verges, industrial basins and quarry lakes; a quarter of the free land of the economic-activity and public-service zones (a judgement on “ne pas mettre en péril la destination première”);

the demonstrable share of the poor-soil land of the agricultural zone (§3.6); and agrivoltaics on permanent and temporary grassland up to 0.6 % of the agricultural area (4 434 ha), the figure of the Livre blanc — which it derives for 2030, whereas the model runs to 2050. Closed are the other farmland, other water bodies, the other unfarmed land of the agricultural zone and the habitat zones. When a policy opens a set of gisements, the park rule is *re-applied to the opened land only*: a strip of verge counts only if it is wide enough on its own, not because the closed field beside it makes the piece large.

3.6 The poor-soil derogation: what can be demonstrated

The circular admits a derogation in the agricultural zone on “des terres où il est démontré que la qualité agronomique du sol est médiocre” (p. 19). The only regional map of agronomic soil quality, *Sols de bonne qualité agronomique* (Pirlot and Degré, ULiège–GxABT, for the AWAC, 2024), is a map of the soil’s water: its five classes are *percentiles* of the plant-available water reserve, corrected for stones, and of the soil depth. Its classes 1–2 (“très faible”, “faible”) cover 110 396 ha; 81 % of them are the stony loams of the Condroz, the Fagne and the Ardenne, and 78 % are declared farmland. They are the bottom of a ranking, not a demonstration: the farmed share of the land a farmer could farm is 91 % and 89 % on classes 1 and 2, 88 % on class 3.

The gisement is the unfarmed land of the agricultural zone mapped in classes 1–2: 4 697 ha after the exclusions and the park rule. The share a permit could realistically claim is built from three independent lines of evidence, cell by cell:

1. **not in the LPIS 2024 agricultural surfaces** — aid-eligible farmland, whether or not declared this year, is “en cours d’exploitation” in the circular’s sense: 2 754 ha remain;
2. **not a garden or a farmyard**: more than 40 m from a built cell: 2 245 ha remain;
3. **revealed agronomic value**: each cell weighs one minus the farmed share of its soil unit (principal CNSW soil type $\times$ AWAC class) over the whole agricultural zone. A soil farmed wherever it is available is not “impropre à la fonction agricole”, whatever its water reserve: 241 ha remain.

The reference opens that share, 5.1 % of the gisement. An independent check agrees: land mapped poor is left unfarmed at 10.0 %, the other soils at 9.6 % — being unfarmed has almost nothing to do with the soil. The bracket is 0 and 100 % (§6); the curtilage distance is priced at 20 and 60 m.

3.7 Energy

Full-load hours come from *atlite* on the model’s 2010 SARAH-3/ERA5 cutout, weighted by where the eligible land lies: 969 h for fixed south-facing parks at the latitude-optimal tilt (the model’s *solar*), 978 h at 25°, 1 038 h for vertical bifacial rows (the east face at its rating plus 0.85 times the west face, which is at the optimistic end of published field results: *atlite* knows no rear-side shading), 953 h for floats at 12° (no cooling gain counted).

4 Calibration against the parks that exist

Wallonia keeps no register of ground-mounted PV parks. OpenStreetMap maps 88 of them of 0.2 ha or more, 241 ha in all — the largest the INEOS Helios park at Jemeppe-sur-Sambre, the former Alconval sandpit at Braine-l’Alleud, Tertre and Burgo Ardennes. Table 1 compares the share of the parks’ land inside each family of exclusions with the share of the Region. 33 % of the parks’ land passes every exclusion, and 72 % once the land-cover test of the “built” family is set aside: WALOUS 2023 is drawn from 2023 imagery, on which a park built earlier is panels, and it classes them as sealed or built surface. The *rules* — the refusals of the circular, the forest and landscape zoning, the hazard maps — are the ones the permitting practice has applied: the parks avoid nature (avoidance 0.03), the forest zone (0.10) and the landscape perimeters (0.05) almost entirely, and slopes moderately (0.57). Most stand in activity zones (33 parks), quarries in

operation (13, the self-supply the circular allows), public-service zones (10), brownfields (7) and landfills (2); 6 stand on unfarmed land of the agricultural zone, by derogation or before 2022.

Table 1: The constraint families against the ground-mounted parks mapped in OSM. Avoidance < 1: the parks avoid the family.

family / rule	Region %	parks %	avoidance	parks mostly inside
nature	14.3	0.4	0.03	1
forest	31.3	3.0	0.10	3
cover	31.9	4.3	0.14	1
landscape	20.0	0.9	0.05	2
hazard	2.5	0.7	0.26	3
terrain	22.2	12.6	0.57	7
built	7.6	44.0	5.80	46
slope>=5%	57.3	35.0	0.61	31
slope>=7%	41.2	25.0	0.61	19
slope>=10%	25.7	15.5	0.60	10
slope>=15%	13.6	8.4	0.62	4
slope>=20%	8.2	5.1	0.62	4

Table 2: The largest parks, their tagged capacity and the gisement they stand on (before exclusions).

park	ha	MWc	MWc/ha	gisement	% eligible
Helios	31.7	60.0	1.89	Economic-activity zones	97
(unnamed)	19.6	31.0	1.58	Other water bodies	0
Parc photovoltaïque de Braine-l’Alleud	15.9	–	–	Landfills (CET)	14
Burgo Ardennes	15.2	9.7	0.64	Economic-activity zones	22
(unnamed)	12.5	–	–	Agricultural zone, not declared	2
Centrale solaire photovoltaïque de Tertre	9.2	6.4	0.70	Habitat zones, unbuilt	4
(unnamed)	7.2	4.6	0.64	Quarries in operation	0
(unnamed)	6.1	–	–	Quarries in operation	6
(unnamed)	5.8	6.0	1.04	Arable crops	90
(unnamed)	5.7	–	–	Brownfields (SAR)	24
(unnamed)	5.4	–	–	Quarries in operation	8
Railpark - Arlon	5.4	1.0	0.19	Other land	46
(unnamed)	5.2	–	–	Quarries in operation	17
(unnamed)	5.0	–	–	Quarries in operation	1
(unnamed)	4.6	–	–	Economic-activity zones	3

The 10 parks of 1 ha or more with a tagged capacity have a median density of 0.97 MWc/ha, 1.25 weighted by area: the fixed-tilt parks of 2020–2024 (Tertre, Burgo Ardennes) are at 0.64–0.70, the two largest and most recent reach 1.6–1.9. The reference density of 1.00 sits at the median of the fleet; the bracket 0.70–1.40 covers the older parks and part of the dense layouts.

5 Results

5.1 How much land each family removes

The forest zone and woodland remove the most land (Table 3, Figure 1): 3 701 and 748 km². Nature takes 2 416, landscape and heritage 855, slope and north-facing slopes 1 100, buildings and networks 934; the natural hazards, after the rest, only 157. The park rule then drops 696 km² of pieces too small or too narrow, leaving 6 275 km², 37.1 % of the Region.

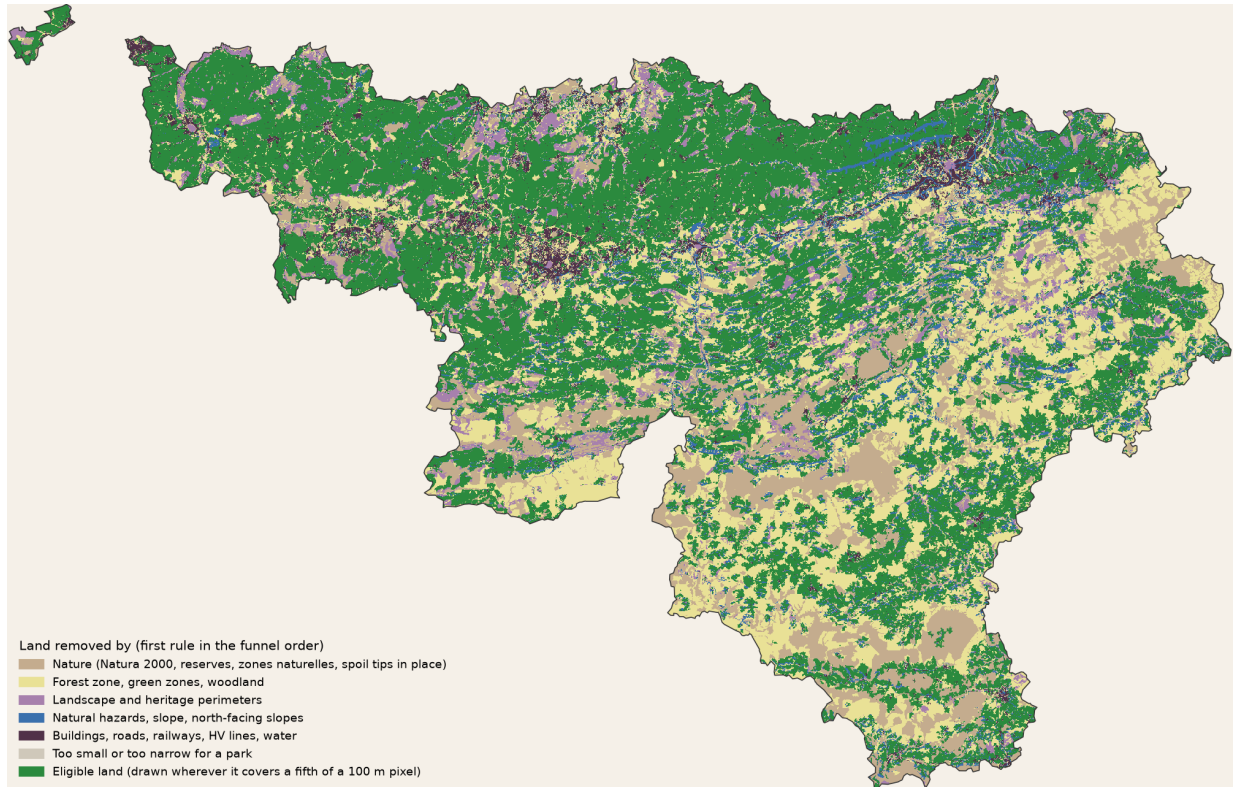


Figure 1: The first family of rules that removes each piece of land, and the land that survives the park rule.

Table 3: Land left after each family, in the order applied.

step	km ² left	% of the Region
Walloon Region	16 905	100
– nature	14 490	85.7
– forest and green zones	10 788	63.8
– woodland (land cover)	10 040	59.4
– landscape and heritage	9 185	54.3
– natural hazards	9 027	53.4
– slope and north-facing slopes	7 927	46.9
– buildings and networks	6 993	41.4
– water	6 971	41.2
– park rule (1 ha, 60 m)	6 275	37.1

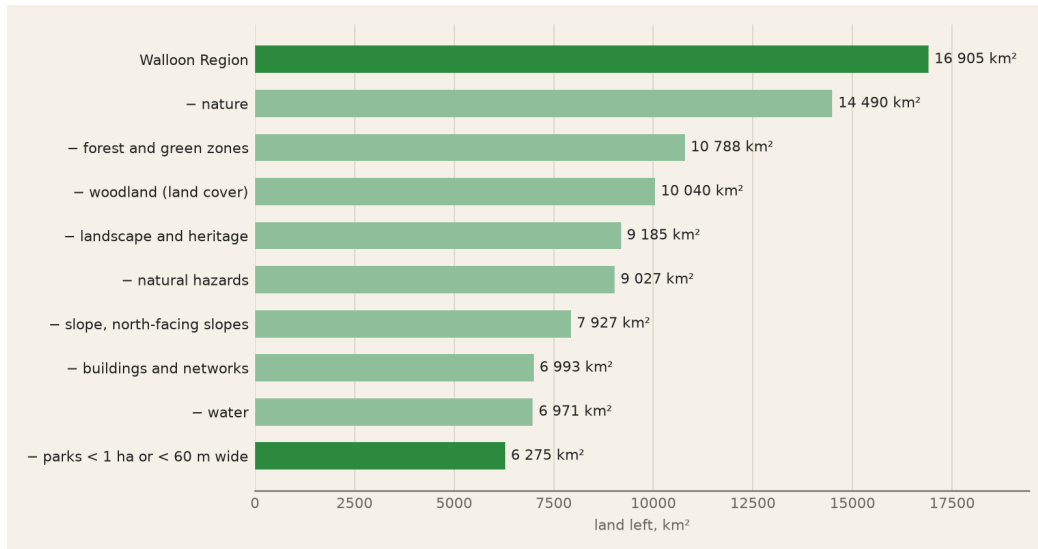


Figure 2: Land left after each family of rules.

5.2 What the land is

Almost all of the land left is farmland (Table 4, Figures 3 and 4): 173 872 ha of permanent grassland, 319 046 ha of arable land. The degraded categories are small by comparison: 967 ha of brownfields after exclusions and the park rule — the SAR perimeters carry buildings, woodland and slopes — 693 ha of unexploited extraction zones, 66 ha of landfills, 949 ha of verges before the park rule is re-applied to them alone (226 ha after), and 544 ha of industrial basins and quarry lakes. The free land of the activity and public-service zones (2 788 and 2 762 ha) is the largest non-agricultural reserve, and a quarter of it is opened.

Table 4: Gisements: eligible land, land after the park rule, technical capacity, and what the reference opens. Open share in %; the reference capacity is computed on the park rule re-applied to the opened land.

gisement	system	eligible ha	after park rule ha	MWc/ha	technical MWc	open %	reference MWc
Industrial basins and quarry lakes	floating	544	544	0.38	204	100	204
Other water bodies	floating	1 514	1 514	0.38	568	0	0
Landfills (CET)	ground	99	66	1.00	66	100	65
Brownfields (SAR)	ground	1 768	967	1.00	967	100	861
Levelled or reworked spoil tips	ground	121	47	1.00	47	100	41
Quarries in operation	ground	1 564	981	1.00	981	0	0
Permanent grassland	agrivoltaic grazing	186 751	173 872	0.70	121 711	100	2 647
Temporary grassland, fodder	agrivoltaic grazing	31 121	30 178	0.70	21 125	100	456
Arable crops	agrivoltaic crops	323 174	319 046	0.35	111 666	0	0
Orchards, vines, small fruit	agrivoltaic canopy	3 175	3 099	0.60	1 860	0	0
Fallow, agri-environmental strips	agrivoltaic grazing	7 378	6 938	0.70	4 856	0	0
Extraction zones (not in operation)	ground	969	693	1.00	693	100	495
Motorway and railway verges	ground	2 728	949	1.00	949	100	226
Economic-activity zones	ground	5 141	2 788	1.00	2 788	25	584
Public-service and leisure zones	ground	5 160	2 762	1.00	2 762	25	535
Agricultural zone, not declared, poor soil	ground	5 281	4 697	1.00	4 697	5.1	140
Agricultural zone, not declared	ground	57 807	50 304	1.00	50 304	0	0
Habitat zones, unbuilt	ground	63 962	29 721	1.00	29 721	0	0
Other land	ground	841	399	1.00	399	0	0
Total		699 096	629 564		356 362		6 254

Finding. The land is not the constraint: 356 GWc of technical potential, of which 261 GWc on declared farmland and 95.1 GWc on everything else. The reference opens 6.3 GWc, about 6.1 TWh a year: 3.2 GWc off the farmland and 3.1 GWc of agrivoltaics.

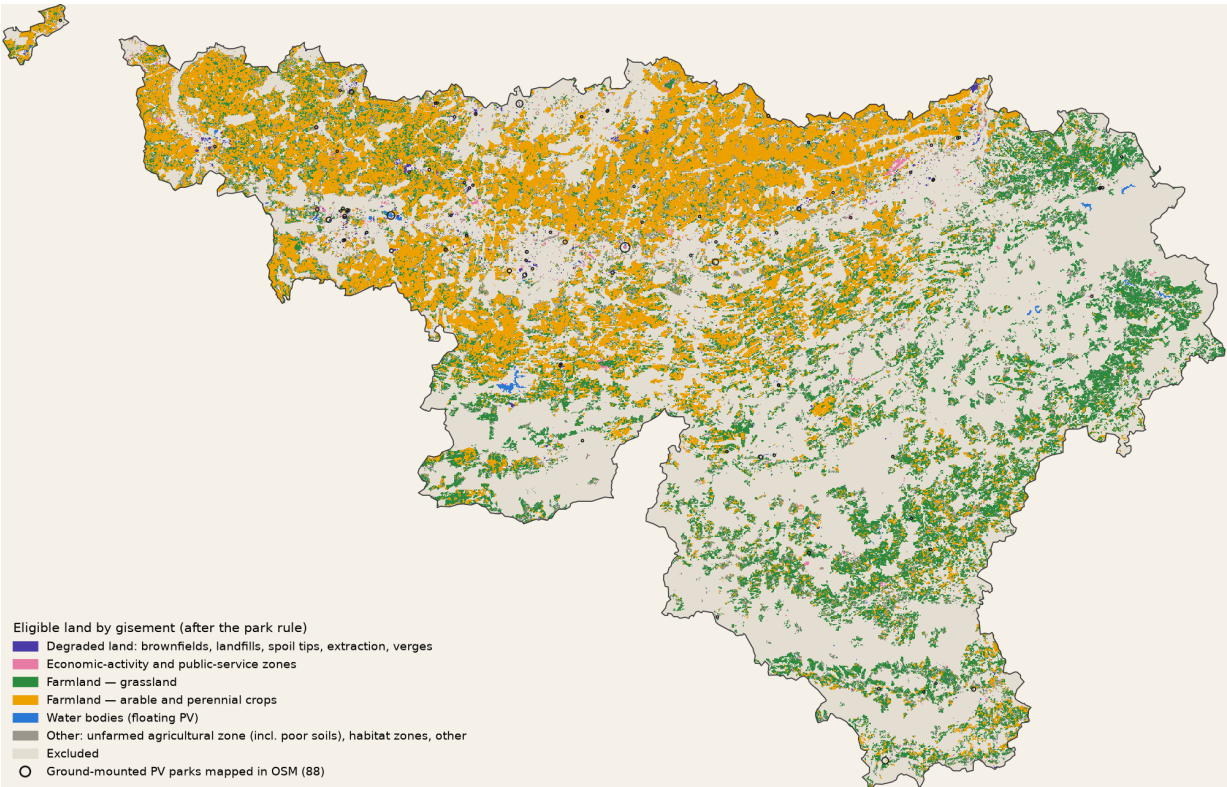


Figure 3: Eligible land after the park rule, by gisement group, and the ground-mounted parks of OSM.

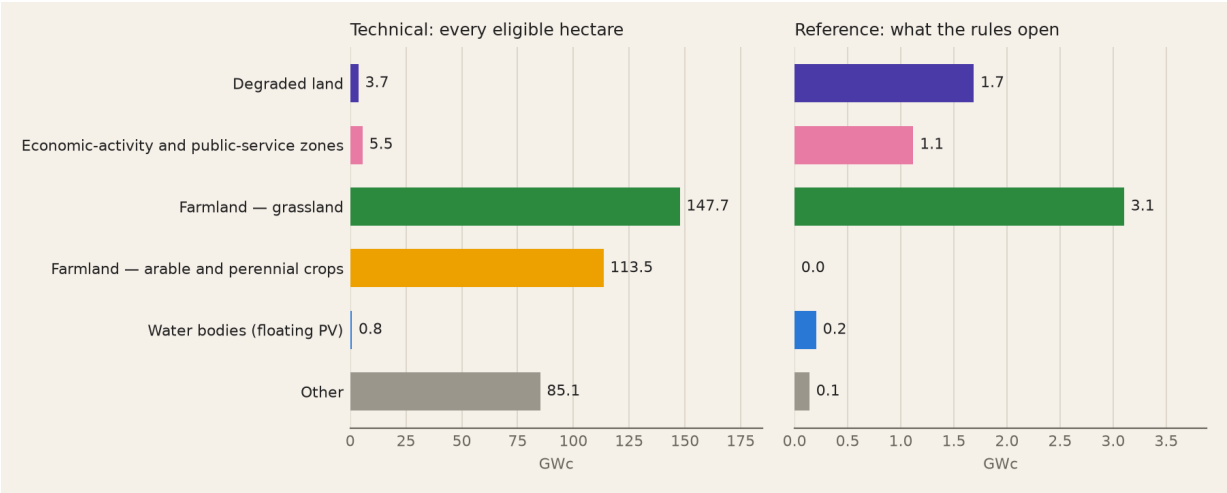


Figure 4: Capacity by gisement group: technical, and what the reference opens.

6 What the answer depends on

Every case is the reference with one change (Table 5, Figure 5).

Table 5: Every choice priced one at a time against the reference (6 254 MWc).

case	MWc	Δ MWc	Δ %
<i>policy</i>			
brownfields kept for reindustrialisation (no PV on SAR)	5 373	−881	−14
no PV in economic-activity and public-service zones	4 995	−1 259	−20
all unbuilt, unfarmed land in economic-activity and public-service zones	9 608	+3 354	+54
floating PV on every water body of 1 ha or more, not only industrial ones	6 822	+568	+9
no derogation on poor soils	6 099	−155	−2
derogation on all mapped-poor, unfarmed land of the agricultural zone	8 855	+2 601	+42
no agrivoltaics (circular as it stands: research pilots only)	2 665	−3 589	−57
agrivoltaics on grassland, capped at 1 % of the SAU	8 323	+2 069	+33
agrivoltaics on grassland, capped at 2 % of the SAU	13 496	+7 242	+116
agrivoltaics on every eligible grassland parcel, no quota	125 972	+119 718	+1 914
agrivoltaics on any crop, poor soils only, no quota	24 656	+18 402	+294
agrivoltaics on every eligible parcel, any crop, no quota	251 964	+245 710	+3 929
the circular strictly as it stands: no agrivoltaics, no derogation on poor soils	2 601	−3 653	−58
<i>judgement</i>			
poor soils: gardens and yards taken as 20 m around buildings	6 270	+16	+0
poor soils: gardens and yards taken as 60 m around buildings	6 237	−17	−0
ecological-link perimeters not excluded	6 256	+2	+0
ADESA landscape perimeters excluded as well	6 026	−228	−4
no landscape or heritage perimeter excluded	6 371	+117	+2
extreme-return-period flood zones excluded as well	5 669	−585	−9
catchment protection zones IIa not excluded	6 269	+15	+0
spoil tips still in place admitted	6 264	+10	+0
<i>technical</i>			
maximum slope 10 %	5 902	−352	−6
maximum slope 20 %	6 440	+186	+3
north-facing slopes excluded from 5 %	6 028	−226	−4
no rule on north-facing slopes	6 614	+360	+6
parks down to 0.5 ha	6 563	+309	+5
parks of 5 ha or more	5 040	−1 214	−19
minimum park width 40 m	7 283	+1 029	+16
minimum park width 100 m	5 081	−1 173	−19
low capacity densities	4 415	−1 839	−29
high capacity densities	7 944	+1 690	+27

case	MWc	Δ MWc	Δ %
floating PV on 15 % of a water body (German rule, WHG §36)	6 173	−81	−1
floating PV on 50 % of a water body	6 458	+204	+3
a cell is built from 10 % building cover	6 083	−171	−3

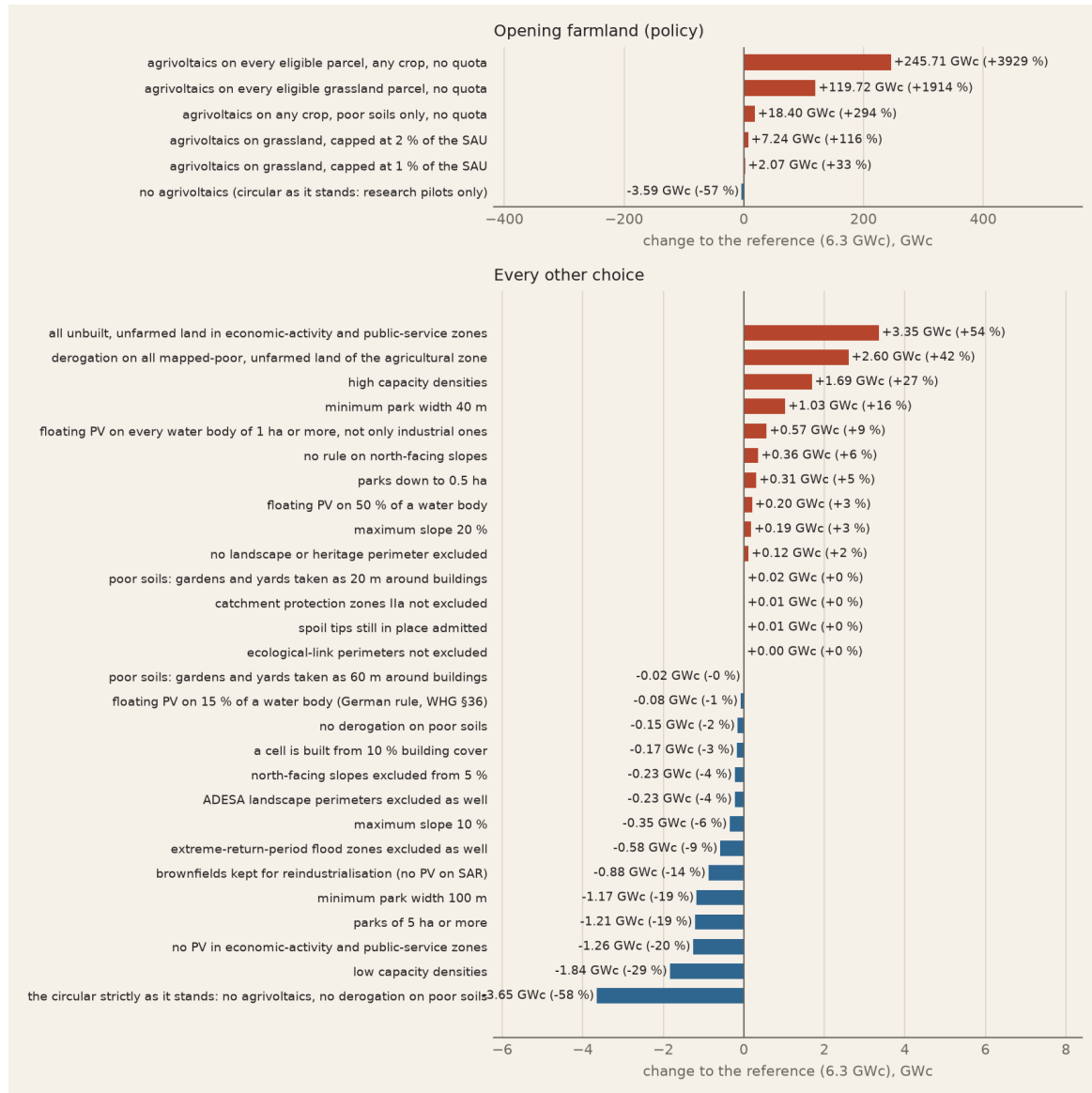


Figure 5: Every choice as a change to the reference. Farmland on its own axis: it is an order of magnitude larger than every other choice.

6.1 Farmland decides

The reference carries agrivoltaics on 4 434 ha, 0.6 % of the agricultural area, 3.1 GWc. Closing them (the circular as it stands: research pilots only) changes the reference by −3.6 GWc; a quota at 1 % gives 8.3 GWc, at 2 % 13.5. Beyond, only a policy choice, not land, sets the answer: every eligible grassland parcel gives 126.0 GWc, every eligible parcel 252.0 GWc. Restricting agrivoltaics to the parcels mapped on poor soils, without quota, gives 24.7 GWc. The Livre blanc’s 0.6 % is

the land the PV still to be built to 2030 would take if all of it were agrivoltaic; for 2050 it is a low reading.

6.2 The other policy choices

Keeping the brownfields for reindustrialisation costs -14% ; closing the activity zones -20% , opening all their free land $+54\%$; floating PV on every water body $+9\%$. On the poor soils, no derogation at all changes the reference by -0.2 GWc and the derogation on all of them by $+2.6$ GWc; the curtilage distance at 20 or 60 m by $+0$ and -0% .

6.3 Judgements and technical criteria

None changes the order of magnitude, but two matter: the park rule (a 5 ha minimum -19% , a 100 m width -19% , 40 m $+16\%$) and the densities. The others are smaller: the landscape perimeters ($+2\%$ if none is excluded), the slope threshold (-6% at 10% , $+3\%$ at 20%), the park size ($+5\%$ down to 0.5 ha, -19% from 5 ha), and the densities ($-29 / +27\%$).

7 Against the other estimates

- **APERe (2020)**: brownfields “ ≥ 2.5 GWc”. This study opens 861 MWc on the SAR inventories after exclusions and the park rule — brownfields carry buildings, woodland and slopes, and a large part of their area is not free.
- **Livre blanc (2024)**: 0.6 % of the agricultural area for the PV still to be built to 2030 — the reference’s agrivoltaic quota; the GLAES areas (56 787 to 245 233 ha) are of the same order as this study’s eligible grassland, and the quota, not the land, is what binds.
- **PACE 2030**: 5 100 GWh of PV, about 6 GWc, all segments together, against 2.4 GWc installed at the end of 2025, mostly on roofs. The ground-mounted reference, 6.3 GWc, is of the same order as the whole 2030 increment.
- **PyPSA-Eur**: 37 GW, a land bound that admits farmland and ignores the plan de secteur.

8 What this means for p_nom_max

The model’s 13 GW is not contradicted by land: 356 GWc of technical potential remain. It is a policy assumption — in effect, the reference with agrivoltaics on 2 % of the agricultural area rather than 0.6 % — and should be documented as such rather than as the sum of two estimates. The workflow gives the number any such assumption implies:

assumption	GWc
circular strictly as it stands (no derogation, no agrivoltaics)	2.6
reference : + demonstrable poor soils, agrivoltaics 0.6 % of the SAU	6.3
reference with agrivoltaics at 1 % of the SAU	8.3
reference with agrivoltaics at 2 % of the SAU	13.5
model cap today	13

A scenario dimension on the farmland rule, rather than one cap, would make the choice visible in the model’s results. No PyPSA-Wal input has been modified.

9 Hypotheses and limits

- **Not represented**: the SGIB (not published as a layer), the *périmètres de point de vue remarquable* (published empty), the grid capacity, the BDES (restricted licence).

- **The activity-zone share** (a quarter) is a judgement: the circular gives no figure, and the free land of an activity zone is also its reserve for firms.
- **Brownfields** are counted as open although the circular admits PV there only temporarily; a temporary park still counts for a 2050 capacity.
- **WALOUS 2023** decides what is built and what is woodland at 1 m; its classification errors (young plantations, glasshouses) pass into the result at the 20 m cell.
- **The OSM inventory** is incomplete and is used for calibration only.

10 Data provenance

dataset	service / layer	features	retrieved
Plan de secteur — zones d’affectation	AMENAGEMENT_TERRI-TOIRE/PDS/22	43 797	2026-09-29
Plan de secteur — réseau routier	AMENAGEMENT_TERRI-TOIRE/PDS/9	5 420	2026-09-29
Plan de secteur — réseau ferroviaire	AMENAGEMENT_TERRI-TOIRE/PDS/11	911	2026-09-29
Plan de secteur — lignes électriques haute tension	AMENAGEMENT_TERRI-TOIRE/PDS/12	1 016	2026-09-29
Plan de secteur — périmètres d’intérêt paysager	AMENAGEMENT_TERRI-TOIRE/PDS/17	1 456	2026-09-29
Plan de secteur — périmètres d’intérêt culturel, historique ou esthétique	AMENAGEMENT_TERRI-TOIRE/PDS/18	771	2026-09-29
Plan de secteur — périmètres de liaison écologique	AMENAGEMENT_TERRI-TOIRE/PDS/19	6	2026-09-29
Périmètres d’intérêt paysager de l’inventaire ADESA	AMENAGEMENT_TERRI-TOIRE/ADESA/1	1 142	2026-09-29
Réseau Natura 2000 — périmètres des sites en vigueur	FAUNE_FLORE/NATURA2000/9	240	2026-09-29
Réserves forestières	FAUNE_FLORE/CONSNAT/1	20	2026-09-29
Réserves naturelles domaniales	FAUNE_FLORE/CONSNAT/2	306	2026-09-29
Réserves naturelles agréées	FAUNE_FLORE/CONSNAT/3	264	2026-09-29
Zones humides d’intérêt biologique	FAUNE_FLORE/CONSNAT/4	73	2026-09-29
Cavités souterraines d’intérêt scientifique	FAUNE_FLORE/CONSNAT/0	80	2026-09-29
Biens classés — sites classés	AMENAGEMENT_TERRI-TOIRE/BC_PAT/1	1 259	2026-09-29
Biens classés — ensembles architecturaux	AMENAGEMENT_TERRI-TOIRE/BC_PAT/2	53	2026-09-29
Biens classés — zones de protection	AMENAGEMENT_TERRI-TOIRE/BC_PAT/4	367	2026-09-29
Patrimoine mondial UNESCO — zones tampon	AMENAGEMENT_TERRI-TOIRE/PAT_MND_UNESCO/1	24	2026-09-29
Zones de contraintes karstiques	AMENAGEMENT_TERRI-TOIRE/CONTR_KARST/0	449	2026-09-29
Contraintes physiques liées aux éboulements	AMENAGEMENT_TERRI-TOIRE/RISQ_EBOULT/0	997	2026-09-29
Versants supérieurs à 30 degrés	AMENAGEMENT_TERRI-TOIRE/RISQ_EBOULT/1	2 834	2026-09-29

dataset	service / layer	features	retrieved
Aléa d'inondation élevé — débordement (et ruissellement associé)	EAU/ALEA_INOND/2	300 882	2026-09-29
Zones inondables — débordement, période de retour extrême	EAU/ZI/2	39 893	2026-09-29
Zones de prévention rapprochée (IIa) des captages	EAU/PROTECT_CAPT/2	736	2026-09-29
Sites à réaménager de droit (SAR)	AMENAGEMENT_TERRI-TOIRE/SAR/0	1 097	2026-09-29
Inventaire des sites à réaménager (SAR de fait)	AMENAGEMENT_TERRI-TOIRE/ISA/4	1 939	2026-09-29
Terrils (inventaire 2018)	SOL_SOUS_SOL/TERRILS_-2018/1	543	2026-09-29
Surfaces occupées par les carrières en activité (2010–2022)	SOL_SOUS_SOL/SURFACES_CARRIERES_ACTIVES/1	155	2026-09-29
PICC — hydrographie, emprise des plans d'eau	TOPOGRAPHIE/PICC_VDIFF/30	39 912	2026-09-29
Carte des sols — principaux types de sols (CNSW simplifiée)	SOL_SOUS_SOL/CNSW__PRINC_-TYPES_SOL/0	200 483	2026-09-29
SIGEC 2024 parcels (bulk GeoPackage)	SPW, CC-BY		2026-09-29
WALOUS 2023 land cover, 1 m	SPW, CC-BY		2026-09-29
LiDAR terrain model 2021–22, 1 m	SPW, CC-BY		2026-09-29
Soil agronomic quality (AWAC/GxABT)	SPW, CC-BY		2026-09-29
Ground-mounted PV parks, substations	OpenStreetMap, ODbL	88	2026-09-29